

$$\varphi_{j,n}(x) = 2^{-j/2} \mathbf{1}_{[2^j n, 2^j(n+1))}(x)$$

$$\psi_{j,n}(x) = 2^{-j/2} \left( \mathbf{1}_{[2^j n, 2^j(n+1/2))}(x) - \mathbf{1}_{[2^j(n+1/2), 2^j(n+1))}(x) \right)$$